JEE Mains 2019 Chapter wise Question Bank

Set and Relations – Questions

Q1

Let Z be the set of integers. If $A = \{x \in Z : 2^{(x+2)(x^2-5x+6)} = 1\}$ and $B = \{x \in Z : -3 < 2x - 1 < 9\}$, then the number of subsets of the set $A \times B$, is :

(1) 2^{15} (2) 2^{18}
(3) 2^{12} (4) 2^{10}

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Q2

Two newspapers A and B are published in a city. It is known that 25% of the city population reads A and 20% reads B while 8% reads both A and B. Further, 30% of those who read A but not B look into advertisements and 40% of those who read B but not A also look into advertisements, while 50% of those who read both A and B look into advertisements. Then the percentage of the population who look into advertisements is:

(1) 13.9 (2) 12.8 (3) 13 (4) 13.5

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Q3

Let A, B and C be sets such that $\phi \neq A \cap B \subseteq C$. Then which of the following statements is not true ?

(1) $B \cap C \neq \phi$

(2) If $(A - B) \subseteq C$, then $A \subseteq C$

(3) $(C \cup A) \cap (C \cup B) = C$

(4) If $(A - C) \subseteq B$, then $A \subseteq B$

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Set and Relations - Solutions

Q1

(1) Let $x \in A$, then

$\because \quad 2^{(x+2)(x^2-5x+6)} = 1 \Rightarrow (x+2)(x-2)(x-3) = 0$

$x = -2, 2, 3$

$A = \{-2, 2, 3\}$

Then, $n(A) = 3$

Let $x \in B$, then

$-3 < 2x - 1 < 9$

$-1 < x < 5$ and $x \in Z$

$\therefore \quad B = \{0, 1, 2, 3, 4\}$

$n(B) = 5$

$n(A \times B) = 3 \times 5 = 15$

Hence, Number of subsets of $A \times B = 2^{15}$

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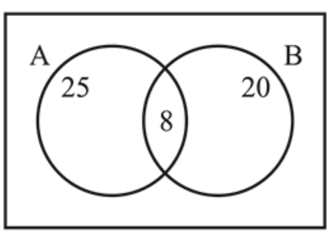
Q2

(1)



% of people who reads A only = 25 – 8 = 17%

% of people who read B only = 20 – 8 = 12%

% of people from A only who read advertisement

$= 17 \times 0.3 = 5.1\%$

% of people from B only who read advertisement

$= 12 \times 0.4 = 4.8\%$

% of people from A & B both who read advertisement

$= 8 \times 0.5 = 4\%$

$\therefore$ total % of people who read advertisement

$= 5.1 + 4.8 + 4 = 13.9\%$

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Q3

(4) (1), (2) and (4) are always correct.

In (3) option,

If $A = C$ then $A - C = \phi$

Clearly, $\phi \subseteq B$ but $A \subseteq B$ is not always true.

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